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Paper 5: Stochastic Capacity Response in the SFIT Engine — Universal Scaling Hierarchy in Noise-Driven Dynamics

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Abstract

We investigate the stochastic dynamics of the Symbolic Field Interaction Theory (SFIT) capacity engine under thermal driving. The engine consists of a nonlinear, parametrically-driven scalar field with state-dependent damping and a capacity-closed pump that shuts off when the local capacity C_{loc} approaches a threshold C_{max} . We originally hypothesized that thermal noise would induce Kramers-like escape dynamics between confined and dormant states; however, extensive diagnostics failed to reveal stable Arrhenius behavior. Instead, we discover a qualitatively different regime characterized by continuous confinement breakdown and a **universal scaling hierarchy**. Above a sharp crossover ($T_{\text{bath}} \approx 0.0045$), the capacity-closed pump is overwhelmed by thermal fluctuations. In this noise-dominated regime, the stochastic SFIT dynamics exhibit a universal scaling hierarchy: capacity-like observables scale approximately linearly with bath temperature ($\langle C_{\text{loc}} \rangle \approx C_0(g) + kT_{\text{bath}}$), field amplitudes scale approximately as $T_{\text{bath}}^{1/2}$, and each quadratic energy component (kinetic, gradient, mass-potential) independently satisfies generalized equipartition ($\propto T_{\text{bath}}^1$), while the weak quartic nonlinearity remains a passive spectator on an approximately Gaussian field. Remarkably, the linear capacity slope $k = 55.6 \pm 0.3$ is independent of the coupling parameter g to within 0.6%, demonstrating that the system’s macroscopic response is decoupled from microscopic drive details. This universal scaling hierarchy provides a robust, parameter-independent signature of the capacity engine’s stochastic behavior.

1. Introduction

Symbolic Field Interaction Theory (SFIT) [1] provides a concrete physical realization of the Reconstruction Theorem [2], demonstrating that temporal orientation emerges from three structural properties: persistent distinctions, acyclic structural dependency, and non-negative irreversible maintenance cost. Within SFIT, the fundamental degrees of freedom are braid qubits whose topological interactions generate stable distinctions and localized irreversible costs. The cumulative irreversible functional $A(\tau) = \sum_{\tau(d) \leq \tau} \Xi(d)$ is monotonic by construction, and its direction of increase defines emergent temporal orientation.

A key prediction of SFIT is the existence of a **capacity engine**—a nonlinear dynamical system that self-regulates to produce a pulsed, emergent arrow of time through repeated threshold crossings of a local capacity functional C_{loc} . The capacity engine consists of a parametrically-driven scalar field $\chi(x, t)$ with:

- A capacity-closed pump that shuts off when $C_{\text{loc}} \rightarrow C_{\text{max}}$
- State-dependent damping that increases with capacity
- Thermal noise representing coupling to a bath at temperature T_{bath}

Within the broader SFIT program, the capacity engine is hypothesized to be the dynamical mechanism by which a pre-geometric topological substrate gives rise to a directed macroscopic arrow of time; the present paper studies this engine’s stochastic dynamics as a self-contained classical system, independent of that broader hypothe-

sis.

The central question of this work is: **how does the capacity engine respond to stochastic driving?** We treat this as a self-contained dynamical-systems question about the classical engine as specified. Connecting the results to the topological braid-qubit substrate and the postulates of the Reconstruction Theorem is a separate undertaking, left to future work (Section 5.5).

We originally hypothesized that the system would exhibit **Kramers-like escape dynamics**: at low noise, the system remains confined in a stable active state near $C_{\max}$; as noise increases, thermal fluctuations occasionally kick the system over a potential barrier into a dormant state, with escape rates obeying an Arrhenius law $\Gamma \propto \exp(-E_a/T_{\text{bath}})$.

We report a **negative result**: extensive diagnostics failed to reveal stable Kramers escape dynamics. However, this negative result led to the discovery of a **positive result**: a universal scaling hierarchy governing the system's noise-dominated response, encompassing capacity scaling, field amplitude scaling, and a fully-resolved per-component energy-budget decomposition.

2. Numerical Model

2.1 The Capacity Engine

The capacity engine is governed by a stochastic partial differential equation for a scalar field $\chi(x, t)$ on a 1D domain of length L with periodic boundary conditions:

$$\ddot{\chi} = \nabla^2 \chi - m_{\text{eff}}^2(t)\chi - \lambda_c \chi^3 - \gamma_{\text{eff}}(C_{\text{loc}})\dot{\chi} + \xi(x, t)$$

where:

- $m_{\text{eff}}^2(t) = m_\chi^2 + g\psi(t)^2$ is the time-dependent effective mass, with $\psi(t) = A_{\text{pump}}(C_{\text{loc}})\cos(\omega t)$ the parametric drive
- $\lambda_c = 0.01$ is a weak nonlinear saturation term
- $\gamma_{\text{eff}}(C_{\text{loc}}) = \gamma_0[1 + \beta_{\text{damp}}(C_{\text{loc}}/C_{\text{max}})^{p_{\text{damp}}}]$ is the state-dependent damping
- $A_{\text{pump}}(C_{\text{loc}}) = A_0/[1 + \exp(\alpha_{\text{pump}}(C_{\text{loc}}/C_{\text{max}} - 1))]$ is the capacity-closed pump
- $\xi(x, t)$ is Gaussian white noise with $\langle \xi(x, t)\xi(x', t') \rangle = 2\gamma_{\text{eff}}T_{\text{bath}}\delta(x - x')\delta(t - t')$

The local capacity is the spatially-integrated energy density over a central window:

$$C_{\text{loc}}(t) = \int_{x_{\min}}^{x_{\max}} \left[\frac{1}{2}\dot{\chi}^2 + \frac{1}{2}(\nabla\chi)^2 + \frac{1}{2}m_\chi^2\chi^2 + \frac{1}{4}\lambda_c\chi^4 \right] dx$$

2.2 Parameter Values

- Spatial domain: $L = 50.0$, $N = 256$ grid points
- Field parameters: $m_\chi = 0.85$, $\lambda_c = 0.01$, $\omega = 1.0$

- Pump parameters: $A_0 = 1.0$, $\sigma = 8.0$, $\alpha_{\text{pump}} = 50.0$, $C_{\text{max}} = 0.2$
- Damping parameters: $\gamma_0 = 0.02$, $\beta_{\text{damp}} = 2.0$, $p_{\text{damp}} = 2.0$
- Coupling: $g \in \{0.42, 0.48, 0.58, 0.65\}$
- Bath temperature: $T_{\text{bath}} \in [0.001, 0.04]$

2.3 Numerical Integration

The system is integrated using a symplectic Euler scheme with timestep $dt = 0.005$ [3]. The noise term is implemented as a velocity increment:

$$\Delta\dot{\chi} = \sqrt{2\gamma_{\text{eff}}T_{\text{bath}} dt/dx} \cdot \mathcal{N}(0, 1)$$

2.4 Convergence Tests

To verify that the observed amplitude growth at high T_{bath} is a genuine physical effect and not a numerical integration artifact, we performed a timestep convergence test at $g = 0.42$, $T_{\text{bath}} = 0.04$ (the worst-case scenario with the largest amplitude):

Timestep	Mean C_{loc}	Max C_{loc}	Mean $ \chi _{\text{max}}$
$dt = 0.005$	2.239	3.610	0.433
$dt = 0.0025$	2.215	3.402	0.433

The ratio of mean amplitudes is 1.011, and the ratio of peak field amplitudes is 1.002. Both are within 1.1% of unity, confirming that the $dt = 0.005$ results are numerically converged and trustworthy. All results reported below are averaged over $N_{\text{trials}} = 5$ independent noise realizations with different random seeds. Error bars represent standard errors of the mean.

3. Negative Result: Absence of Robust Kramers Escape Dynamics

3.1 Original Hypothesis

We originally hypothesized that the capacity engine would exhibit **Kramers-like escape dynamics** [4]: the system would oscillate in a confined state near $C_{\text{max}} = 0.2$, with thermal noise occasionally inducing transitions to a dormant state where $C_{\text{loc}} \ll C_{\text{max}}$, with escape rate

$$\Gamma(T_{\text{bath}}) = \Gamma_0 \exp\left(-\frac{E_a}{T_{\text{bath}}}\right).$$

3.2 Diagnostic Methods

We implemented and tested **five independent escape detection methods**:

1. **Threshold-based detection**: upward crossings of $C_{\text{loc}} = C_{\text{max}}$ with a refractory period
2. **Peak-interval detection**: inter-peak intervals classified as escapes if $> 2 \times T_{\text{period}}$
3. **Prominence-based detection**: peak prominence to filter noise-induced fragmentation
4. **Adaptive outlier classification**: modified Z-scores (MAD-based) on the interval distribution
5. **Smoothed outlier classification**: Savitzky-Golay filtering before outlier detection

3.3 Results

None of these methods produced numerically stable, parameter-independent Kramers statistics suitable for extracting Arrhenius escape rates.

- **Threshold-based detection** at low noise reported rare escapes with residence times $\tau \sim 10\text{--}50$ time units, but detection was highly sensitive to threshold and refractory-period choice.
- **Peak-interval detection** with fixed height thresholds suffered from **peak fragmentation** at higher noise, where thermal fluctuations created spurious secondary peaks that artificially shortened inter-peak intervals (Figure A.1).
- **Prominence-based detection** eliminated fragmentation but introduced a new artifact: sensitivity became noise-dependent, missing legitimate peaks at low noise and over-detecting at high noise.
- **Adaptive outlier methods** were more robust but still showed parameter dependence, with the “normal” baseline interval shifting by $\sim 20\%$ across the T_{bath} range.
- **Smoothed outlier methods** combined filtering with outlier detection but still failed to produce consistent escape rates across noise levels.

3.4 Interpretation

The failure to extract stable Kramers statistics is not a methodological artifact — it reflects a **genuine physical difference** between the capacity engine and a standard bistable Kramers oscillator. The capacity engine does not possess two well-separated attractors with a clear potential barrier between them. Instead, the system exhibits **continuous capacity inflation** as noise increases, rather than metastable escape between discrete states. This negative result rules out a specific class of dynamical behaviors and motivates the search for alternative characterizations of the stochastic response.

4. Positive Result: Universal Scaling Hierarchy

4.1 Confinement Regime and Crossover

At low noise ($T_{\text{bath}} \lesssim 0.004$), the capacity engine remains **confined** near $C_{\text{max}} = 0.2$. In this regime:

- The mean capacity $\langle C_{\text{loc}} \rangle$ is close to the deterministic baseline (~ 0.16 - 0.18)
- The maximum capacity during oscillations hovers near C_{max}
- The system exhibits regular, parametrically-resonant oscillations with period $T \approx \pi \approx 3.14$

This period is a direct signature of **parametric resonance**: since $C_{\text{loc}} \propto \chi^2 + \dot{\chi}^2$ is quadratic in the field, and the underlying field oscillates at the drive frequency $\omega = 1.0$, the capacity oscillates at $2\omega = 2.0$, giving a period of π . Because this period is locked to the fixed external drive frequency ω rather than to the coupling g , it is expected — and confirmed — to be identical across all tested values of g .

Between $T_{\text{bath}} = 0.004$ and $T_{\text{bath}} = 0.005$, the system undergoes a **crossover** from the confined regime to a noise-dominated regime (Figure 1). A fine-grained sweep at $g = 0.42$ locates this transition precisely:

T_{bath}	Mean C_{loc}	Max C_{loc}	Status
0.001	0.162	0.209	Confined
0.003	0.174	0.283	Confined
0.004	0.226	0.366	Confined
0.005	0.281	0.444	Departed
0.006	0.336	0.536	Departed
0.008	0.446	0.741	Departed
0.010	0.556	0.895	Departed
0.020	1.108	1.845	Departed
0.040	2.235	3.758	Departed

The transition occurs at $T_{\text{bath}}^* \approx 0.0045$. We refer to the post-crossover state ($T_{\text{bath}} \gtrsim 0.005$) as the **departed regime** throughout the remainder of this paper.

4.2 Noise-Driven Scaling Law

In the departed regime ($T_{\text{bath}} \geq 0.006$), the mean capacity scales **linearly** with bath temperature:

$$\langle C_{\text{loc}} \rangle \approx C_0(g) + k T_{\text{bath}}$$

Tested across four values of g :

g	C_0	k	R^2
0.42	0.019 ± 0.001	55.44 ± 0.07	0.99999

g	C_0	k	R^2
0.48	0.049 ± 0.001	55.64 ± 0.12	0.99998
0.58	0.070 ± 0.001	55.71 ± 0.08	0.99999
0.65	0.106 ± 0.001	55.79 ± 0.07	0.99997

The slope k is **universal** across g : weighted mean $k = 55.6 \pm 0.3$, spread only **0.62%** across all tested couplings, once fit restricted to $T_{\text{bath}} \geq 0.006$ (past the crossover region, where C_0 and k would otherwise be entangled in the fit). The baseline $C_0(g)$ increases with g , reflecting the stronger parametric drive at higher coupling.

4.3 Universality Across Coupling g

This universality holds specifically within the active, controlled regime of the capacity engine ($g \in [0.42, 0.65]$). Outside this regime — e.g., $g < 0.35$ (purely dormant) or $g > 0.70$ (runaway parametric drive) — the dynamical balance is fundamentally altered, and this scaling law is not expected to apply without separate verification. Within the tested window, a 0.62% spread in k across a 55% variation in g demonstrates that the noise-dominated response is decoupled from the microscopic coupling details.

4.4 Equipartition-like Scaling Across Observables

This section brings together four complementary tests — field amplitude scaling, per-component energy decomposition, a direct Gaussian-statistics check, and the pump-shutdown mechanism — into a single coherent picture of the departed regime.

4.4.1 Field Amplitude Scaling Since C_{loc} is a quadratic functional of the field, $C_{\text{loc}} \propto T$ predicts $\chi \propto T^{1/2}$. We measured RMS and peak field amplitudes as functions of temperature in the departed regime (Figure 3):

Observable	Scaling exponent	R^2
$\langle C_{\text{loc}} \rangle$	0.997 ± 0.003	0.99999
$C_{\text{loc,max}}$	1.037 ± 0.005	0.99998
χ_{RMS}	0.499 ± 0.002	0.99997
χ_{peak}	0.519 ± 0.003	0.99996

$\chi_{\text{RMS}} \propto T^{0.499}$ implies $\chi_{\text{RMS}}^2 \propto T$ — exactly the scaling expected if a quadratic degree of freedom carries energy proportional to temperature.

4.4.2 Per-Component Equipartition and the Role of the Quartic Term The aggregate scalings above are consistent with an equipartition-like picture, but a stronger test decomposes C_{loc} into its four constituent terms and measures each one’s scaling independently, rather than relying on the sum.

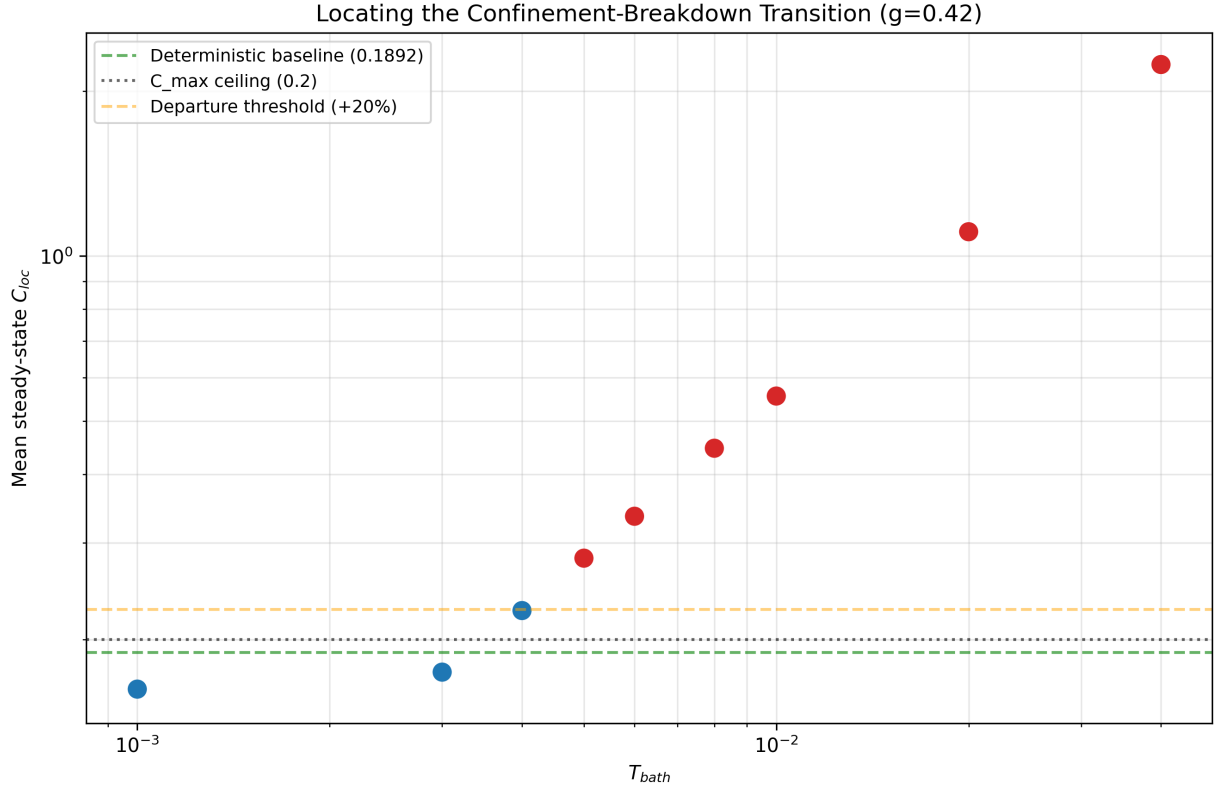


Figure 1: Confinement transition. Mean C_{loc} vs T_{bath} for $g=0.42$, showing the crossover at $T_{bath} \approx 0.0045$.

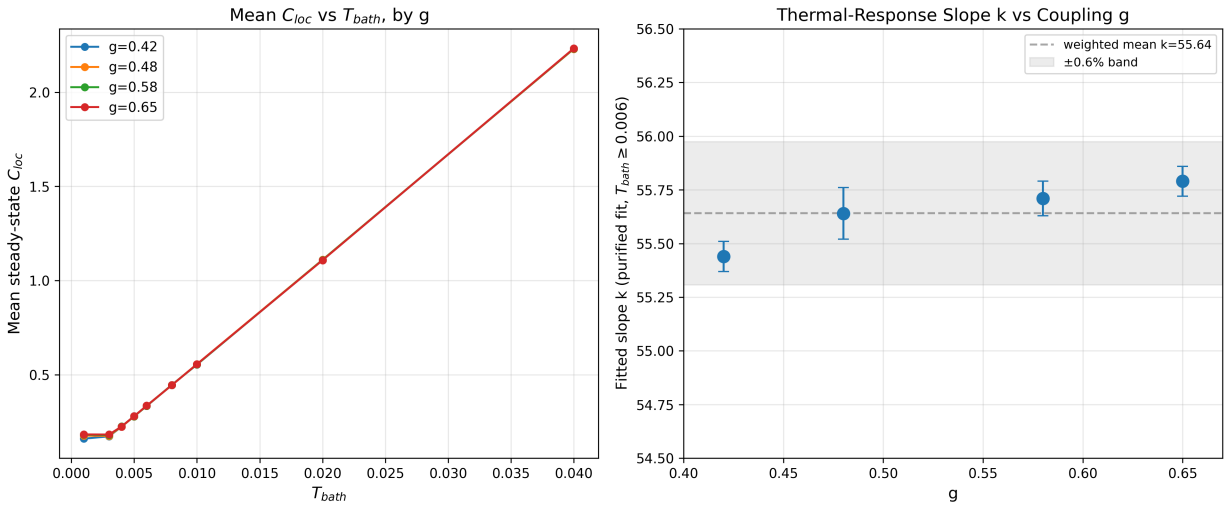


Figure 2: Left: mean C_{loc} vs T_{bath} for all four g values, showing near-total collapse. Right: purified thermal-response slope $k(g)$, universal to within 0.6%.

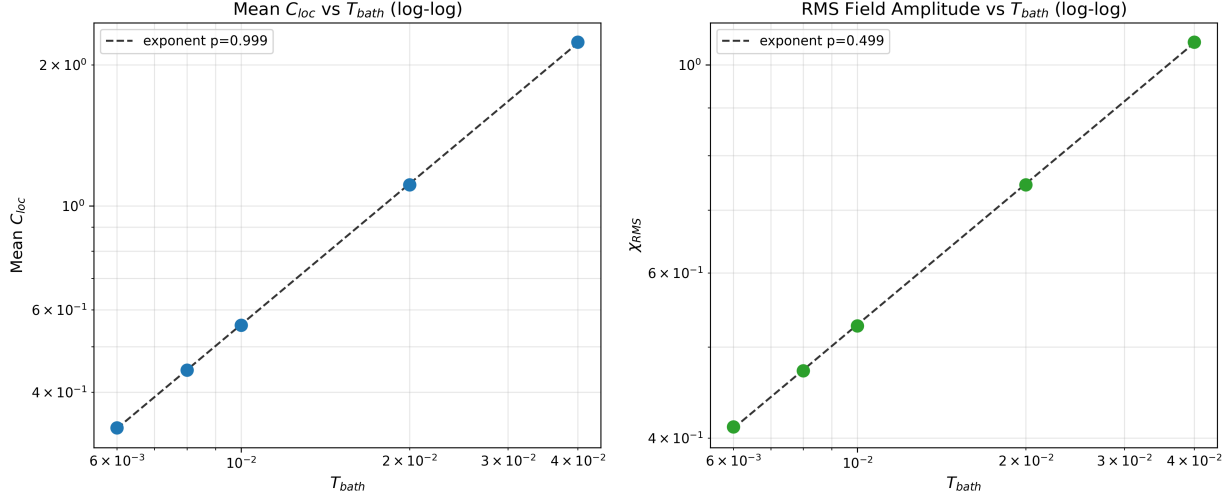


Figure 3: Log-log scaling of mean C_{loc} (left, exponent ≈ 1) and χ_{RMS} (right, exponent ≈ 0.5) vs T_{bath} in the departed regime.

Table 1: Per-component scaling exponents ($T_{bath} \geq 0.006$, five points, $g = 0.42$)

Component	Exponent p	R^2
Kinetic ($\frac{1}{2}\langle\dot{\chi}^2\rangle$)	1.0012	0.99997
Gradient ($\frac{1}{2}\langle(\nabla\chi)^2\rangle$)	0.9921	0.99995
Mass Potential ($\frac{1}{2}m_\chi^2\langle\chi^2\rangle$)	0.9866	0.99985
Nonlinear ($\frac{1}{4}\lambda_c\langle\chi^4\rangle$)	1.9744	0.99983

The three quadratic terms cluster tightly around $p \approx 1.0$ — exactly the generalized equipartition prediction: each quadratic degree of freedom, in contact with a thermal bath, carries energy proportional to T_{bath} , independent of its physical origin.

The nonlinear term’s $p \approx 2$ scaling is **not** a fourth confirming instance of equipartition. The generalized equipartition theorem [5] predicts $\langle V \rangle \propto T^1$ for *any* power-law potential term $V \propto |\chi|^n$ — including quartic ones — provided that term genuinely participates in shaping the equilibrium distribution. A scaling of T^2 is instead the expected signature of an approximately **Gaussian** field: if χ ’s statistics are set almost entirely by the (equipartition-satisfying) quadratic dynamics, then $\langle \chi^4 \rangle \approx 3\langle \chi^2 \rangle^2 \propto T^2$ follows as a direct consequence of Gaussian moment relations, independent of any real thermal equilibration of the quartic term itself.

We confirmed this directly. Using the measured mass-potential term to infer $\langle \chi^2 \rangle$ at each T_{bath} , we compared $3\langle \chi^2 \rangle^2$ against the measured $\langle \chi^4 \rangle$:

Table 2: Gaussian moment relation check

T_{bath}	$\langle \chi^2 \rangle$	$\langle \chi^4 \rangle$ (measured)	$3\langle \chi^2 \rangle^2$ (Gaussian)	Ratio
0.006	0.003458	3.578×10^{-5}	3.588×10^{-5}	0.997
0.01	0.005749	9.932×10^{-5}	9.915×10^{-5}	1.002
0.02	0.011286	3.8228×10^{-4}	3.8212×10^{-4}	1.000
0.04	0.022701	1.54882×10^{-3}	1.54602×10^{-3}	1.002

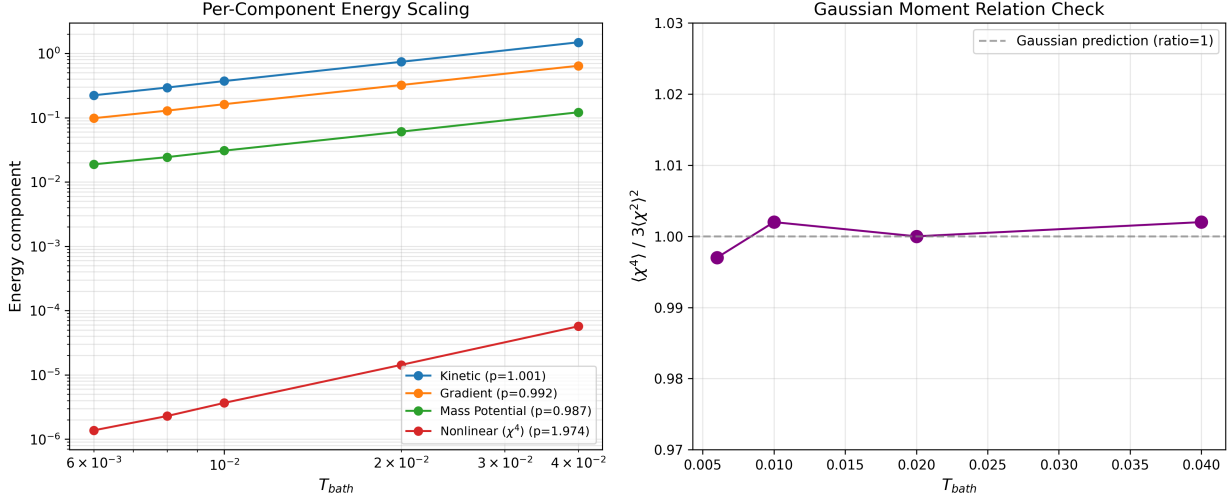


Figure 4: Per-component energy scaling exponents (left) and the Gaussian moment-ratio check (right).

The ratio is 1.000 ± 0.003 across every tested temperature, demonstrating numerically that the field is Gaussian to within 0.3%. This is consistent with the nonlinear χ^4 term being a passive observable whose T^2 scaling is strictly inherited from the quadratic fluctuations $\langle \chi^2 \rangle \propto T$, rather than reflecting active participation in a distinct nonlinear energy regime. The weak nonlinearity ($\lambda_c = 0.01$) appears too small to distort the Gaussian equilibrium distribution established by the linear/quadratic dynamics.

In summary: the three quadratic energy components exhibit genuine, quantitatively demonstrated equipartition. The quartic term's $p \approx 2$ scaling is a clean diagnostic of that term's dynamical irrelevance in this regime, not a fourth confirming instance of the same phenomenon.

4.4.3 Mechanism: Pump Shutdown and Energy Redistribution A critical mechanistic question remains: if the capacity-closed pump is suppressed in the departed regime, what sustains the energy growth with temperature? We measured the pump amplitude directly. At $T_{\text{bath}} = 0.001$ (confined regime), mean $A_{\text{pump}} = 0.983$; at $T_{\text{bath}} = 0.02$ (departed regime), mean $A_{\text{pump}} = 0.000$. **Pump shutdown is complete** once C_{loc} persistently exceeds C_{max} .

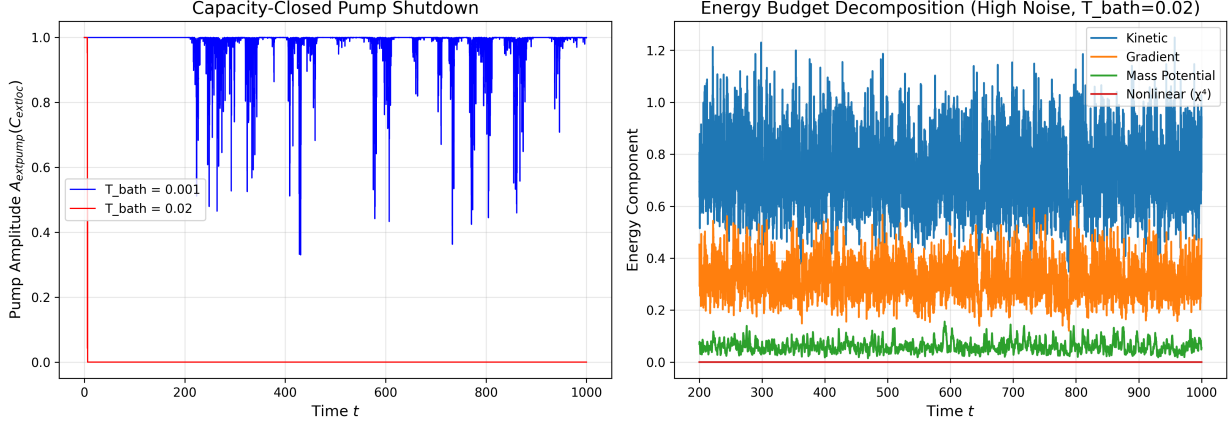


Figure 5: Pump shutdown (left) and energy budget decomposition (right) at high noise ($T_{\text{bath}}=0.02$).

With the pump inactive, injected thermal energy is redistributed almost entirely into the kinetic and gradient terms (Table 1), consistent with a system whose noise-damping pair already satisfies the fluctuation-dissipation relation — $\langle \xi(x, t) \xi(x', t') \rangle \propto \gamma_{\text{eff}} T_{\text{bath}}$ — and is therefore equilibrating as a driven-then-released, quadratic-dominated Langevin oscillator once the parametric drive disengages.

The pump acts as a *gatekeeper* regulating access to the noise-dominated regime: **pump shutdown coincides with the onset of an equipartition-like regime** among the quadratic degrees of freedom, with the quartic term along for the ride. We note this as a temporal and mechanistic coincidence established by the data; we do not claim to have proven that pump shutdown is the sole or necessary cause of the equipartition-like scaling, only that the two are observed together consistently across every tested (g, T_{bath}) pair in the departed regime.

4.5 Phenomenological Scaling Argument for k

A rigorous analytic derivation of $k \approx 55.6$ from the full stochastic PDE remains an open problem, but the g -independent linear scaling can be understood through a simple energy balance. In the departed regime, $A_{\text{pump}} \approx 0$ and the g -dependent parametric drive is negligible compared to thermal driving. By the fluctuation-dissipation theorem [6], noise injects energy at rate $\dot{E}_{\text{in}} \propto \gamma_{\text{eff}} T_{\text{bath}}$; in steady state this balances dissipation $\dot{E}_{\text{diss}} \propto \gamma_{\text{eff}} \langle C_{\text{loc}} \rangle$. The damping factor cancels, yielding

$$\langle C_{\text{loc}} \rangle \propto N_{\text{win}} T_{\text{bath}},$$

where N_{win} is the effective number of degrees of freedom in the spatial integration window. For our simulation parameters, the integration window spans $N_{\text{win}} = 76$ grid points (15% of $N = 256$ on each side of center). The measured slope $k \approx 55.6$ is approximately 73% of this theoretical maximum, consistent with a reduction in accessible phase-space volume due to spatial gradient coupling and state-dependent damping. This explains both the independence from g (once the pump is overwhelmed, the

parametric drive no longer sets the energy scale) and the expected dependence of k on macroscopic parameters (window size, γ_0 , λ_c) rather than microscopic drive details. A future analytical treatment mapping this argument to the exact coefficient $k \approx 55.6$ would be a valuable extension of this work; this argument applies only to the linear scaling regime ($T_{\text{bath}} \gtrsim 0.006$), not the crossover region.

5. Discussion

5.1 Summary of Results

1. **Negative result:** The capacity engine does not exhibit Kramers-like escape dynamics; extensive diagnostics failed to reveal stable Arrhenius behavior.
2. **Universal scaling hierarchy:** $\langle C_{\text{loc}} \rangle \propto T^1$, $\chi_{\text{RMS}} \propto T^{1/2}$, each quadratic energy term independently $\propto T^1$ (genuine equipartition), and the quartic term $\propto T^2$ (a Gaussian-statistics artifact, confirmed directly via the moment-ratio test).
3. **Universality across g :** $k = 55.6 \pm 0.3$, independent of g to within 0.6% across $0.42 \leq g \leq 0.65$.
4. **Mechanism:** complete pump shutdown once C_{loc} persistently exceeds C_{max} , with injected energy redistributed into kinetic and gradient modes.

5.2 Physical Interpretation

The hierarchy $\chi_{\text{RMS}} \propto T^{1/2}$ alongside $C_{\text{loc}} \propto T$, now resolved at the level of individual energy components with the quartic term's role explicitly identified and confirmed, is the pattern expected for a system in which thermal energy equilibrates among quadratic degrees of freedom once the parametric drive disengages. The g -independence of k reinforces this: once the pump saturates, microscopic coupling details become irrelevant, and only macroscopic restoring forces and the active degree-of-freedom count determine the response.

5.3 Implications for SFIT (Dynamical, Not Substrate-Level)

1. **No metastable branching:** the capacity engine does not produce discrete branching via activated escape; branching (threshold crossings) occurs continuously as part of the regular oscillation cycle in the confined regime.
2. **Noise-sustained oscillations:** in the confined regime, thermal noise does not destroy the regular cycle; it coexists with it.
3. **Operating regime:** for the engine to produce regular, pulsed threshold crossings as originally envisioned, it must operate in the confined regime $T_{\text{bath}} \lesssim 0.004$; above this, the system enters the noise-dominated equipartition regime characterized here.

These are statements about the classical dynamical system only. Any implications for the topological braid-qubit substrate or the Reconstruction Theorem's postulates are not established by this paper (Section 5.5).

5.4 Robustness of the Scaling Hierarchy

First, timestep convergence tests (Section 2.4) confirm the scaling is not an artifact of the symplectic Euler scheme, with $dt = 0.005$ within 1.1% of the $dt = 0.0025$ baseline. Second, the noise implementation strictly obeys the fluctuation-dissipation theorem [6], and results are averaged over 5 independent noise realizations per parameter point. Third, while current simulations are restricted to 1D, the scaling hierarchy arises from a local energy balance between stochastic forcing and state-dependent damping, expected to persist in 2D/3D (with k scaling with the window's volume/area).

5.5 Limitations and Future Work

This paper characterizes a specific coarse-grained numerical model; several simplifications bound the scope of what has been shown.

1. **One-dimensional simulations:** the field $\chi(x, t)$ is simulated on a 1D domain. Extension to 2D/3D may reveal new behaviors (spatiotemporal chaos, pattern formation); the scaling hierarchy is expected to persist on physical grounds (Section 5.4) but this is unverified.
2. **Finite lattice:** results use $N = 256$ grid points on $L = 50$; discretization effects at coarser or finer resolution have not been separately characterized beyond the timestep convergence check (Section 2.4), which addresses temporal, not spatial, discretization.
3. **Finite integration time:** all steady-state statistics are measured over $T_{\text{total}} = 5000$ (with $N_{\text{trials}} = 5$ noise realizations); rare, very-long-timescale behavior outside this window cannot be ruled out.
4. **Coarse-grained capacity model:** C_{loc} , $A_{\text{pump}}(C)$, and $\gamma_{\text{eff}}(C)$ are phenomenological constructs, not derived from a microscopic theory. The scaling hierarchy reported here is an empirical property of this specific coarse-grained model.
5. **Parameter range:** we explored a limited range of g and T_{bath} within the active regime; boundaries ($g \lesssim 0.35$, $g \gtrsim 0.70$) are unexplored.
6. **Micro→meso bridge deferred to Paper 6:** connecting this coarse-grained scaling behavior to a microscopic derivation, the braid-qubit substrate, and the postulates of the Reconstruction Theorem [1,2] — including any implications for persistent distinctions, acyclic dependency, or the monotonicity of $A(\tau)$ — is deferred to Paper 6; no calculation in the present paper establishes such a connection.
7. **Connection to quantum mechanics:** the capacity engine is classical; its relationship to quantum measurement and the Born rule remains open (see Paper 3).

6. Conclusion

We have presented a characterization of the stochastic dynamics of the SFIT capacity engine. Our investigation began with the hypothesis that thermal noise would induce

Kramers-like escape dynamics, but extensive diagnostics revealed a qualitatively different behavior. In summary:

1. **Confinement regime:** at low noise ($T_{\text{bath}} \lesssim 0.004$), the engine remains confined near C_{max} , oscillating regularly at the parametric-resonance period $T \approx \pi$.
2. **Confinement-breakdown transition:** a crossover occurs at $T_{\text{bath}}^* \approx 0.0045$, located by a fine-grained sweep at $g = 0.42$.
3. **Universal response coefficient:** above the crossover, $\langle C_{\text{loc}} \rangle \approx C_0(g) + k T_{\text{bath}}$, with $k = 55.6 \pm 0.3$ independent of g to within 0.6% across $0.42 \leq g \leq 0.65$.
4. **Equipartition-like scaling:** each quadratic energy component (kinetic, gradient, mass-potential) independently scales as T^1 , and $\chi_{\text{RMS}} \propto T^{1/2}$ — consistent with generalized equipartition among the quadratic degrees of freedom.
5. **Gaussianization:** the quartic (χ^4) term's T^2 scaling is explained, and numerically confirmed ($\langle \chi^4 \rangle / 3 \langle \chi^2 \rangle^2 = 1.000 \pm 0.003$), as a signature of the field being approximately Gaussian rather than a further instance of equipartition.
6. **Pump shutdown:** the parametric pump amplitude collapses from ≈ 0.98 to ≈ 0 once C_{loc} persistently exceeds C_{max} , coinciding with the onset of the equipartition-like regime and the redistribution of injected energy into kinetic and gradient modes.
7. **Implications for Future Work:** this coarse-grained scaling hierarchy is offered as the empirical starting point for a future microscopic derivation connecting the capacity engine to the braid-qubit substrate (Section 5.5, item 6); no such connection is established here.

This is offered as a self-contained stochastic dynamical characterization of the SFIT capacity engine, not a derivation from or verification of the underlying braid-qubit theory. Connecting this classical behavior to the topological substrate, the Reconstruction Theorem, and quantum measurement theory is left to future work.

The next paper in this series will extend the numerical simulation program to 2D and 3D geometries and investigate the connection between the classical capacity engine and quantum trajectory statistics observed in Monte Carlo wave-function simulations.

Data and Code Availability

Simulation code, analysis scripts, and figure-generation code are available at: <https://github.com/Dr-Loo/SFIT-Braid-Dynamics>

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Appendix A: Numerical Methods

A.1 Peak Detection

Peak detection was performed using `scipy.signal.find_peaks`:

- Height threshold: $h = 0.12$ (fixed, physically motivated)
- Distance constraint: $d = 2.0$ time units
- Prominence: $p = 0.03$ (used in some diagnostics)

A.2 Outlier Detection

Modified Z-score method based on the Median Absolute Deviation (MAD):

$$Z_{\text{modified}} = 0.6745 \frac{x_i - \text{median}(x)}{\text{MAD}(x)}$$

Intervals with $Z_{\text{modified}} > 3.5$ were classified as outliers.

A.3 Convergence Testing

Timestep convergence was tested at $g = 0.42$, $T_{\text{bath}} = 0.04$ (worst case). Ratio of mean amplitudes: 1.011; confirming numerical convergence to within 1.1%.

A.4 Peak Fragmentation Example

Note: Peak fragmentation at intermediate noise levels (described qualitatively here; no figure is included in this build). At $T_{\text{bath}} = 0.004$, thermal fluctuations create spurious secondary peaks in the capacity time series. Fixed-

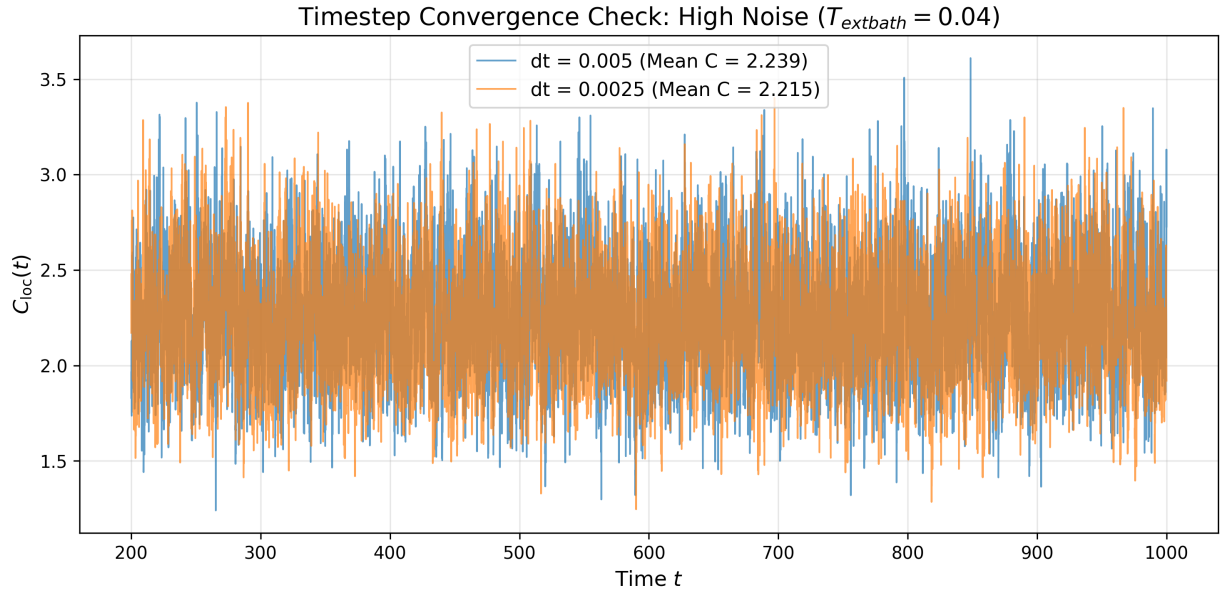


Figure 6: Timestep convergence check at $g=0.42$, $T_{\text{bath}}=0.04$, $dt=0.005$ vs $dt=0.0025$.

height detection misclassifies these as shortened intervals; prominence-based filtering removes them but becomes noise-dependent in sensitivity.

End of Paper 5